

Task 1b – Designing Documents

Low Fidelity Mock ups:

Sign Up Page:

Title

Log In

Sign Up



LOG-IN TITLE

TEXT

TEXT

TEXT

TEXT



TEXT

Sign - up option

Login Page:

Title

Log In

Sign Up



Sign - Up Title

TEXT

TEXT



TEXT

TEXT



TEXT

TEXT

TEXT

LOG - IN OPTION

Home Page:

GLH

HomeShopProducers

My Account



Buy Fresh Local Food from Your Community

Connecting local producers with local people

Search for local produce...

Q

FEATURED PRODUCTS

0 of 0 items

Product Image Placeholder

Organic Carrots

NEW

\$1.50 / kg

Add to Basket

Product Image Placeholder

Free Range Eggs

NEW

\$2.20 / half dozen

Add to Basket

Product Image Placeholder

Sourdough Loaf

LOW STOCK

\$5.50

Add to Basket

NEWS & UPDATES

New Producer Joined

Volunteering Village Farm to our community network this week.

Seasonal Harvest

The first of the season's strawberries are now available for order.

Community Meeting

Join us next Thursday for our monthly producer-customer meeting.

GLH

Connecting local food producers directly with customers in the local community.

About GLH

Contact

Privacy Policy

Terms of Service

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Product Categories:



Search Bar

Search products, categories, or producers...

Filters

Price Range

50

500+

Category

☐ Vegetables

☐ Dairy

☐ Meat

☐ Bakery

Producer

Select Producer ▼

Clear Filters

Product Grid

IMAGE PLACEHOLDER Organic Carrots Green Valley Farm £1.50 Add to IN STOCK	IMAGE PLACEHOLDER Whole Milk 2L Dairy Hill £2.20 Add to IN STOCK	IMAGE PLACEHOLDER Beef Mince 500g Heritage Meats £4.50 Add to LOW STOCK	IMAGE PLACEHOLDER Sourdough Loaf The Village Bakery £3.00 Add to IN STOCK
IMAGE PLACEHOLDER Free Range Eggs Happy Hen £2.80 Add to IN STOCK	IMAGE PLACEHOLDER Red Apples (6) Orchard Direct £2.00 Add to IN STOCK	IMAGE PLACEHOLDER Cheddar Cheese West Country Creamery £3.50 Add to IN STOCK	IMAGE PLACEHOLDER Pork Sausages Heritage Meats £4.00 Add to OUT OF STOCK

Product Detail Page:

GLH Logo

ShopProducersOur StoryContact

Q

Person

Shopping Bag

Product Image

Main Product Image Placeholder

Product Info

ORGANIC DAIRY

Fresh Farm Milk (2L)

IN STOCK

SKU: GLH-002+

£3.50

-1+

ADD TO BASKET

Producer: Green Valley Farm

Delivery:

Origin:

Next Day Available

Somerset, UK

Description

Meet the Producer

Allergens & Traceability

Reviews (2)

Description

This organic whole milk is sourced from grass-fed cows at Green Valley Farm. It is pasteurized but unhomogenized to maintain its natural creamy texture and rich flavor profile. Perfect for coffee, cereal, or drinking chilled.

Producer Story

Green Valley Farm

Green Valley Farm has been family-operated for three generations. They prioritize animal welfare and regenerative farming practices to ensure the highest quality dairy products while protecting the local ecosystem.

Reviews

★★★★☆

4.5 / 5

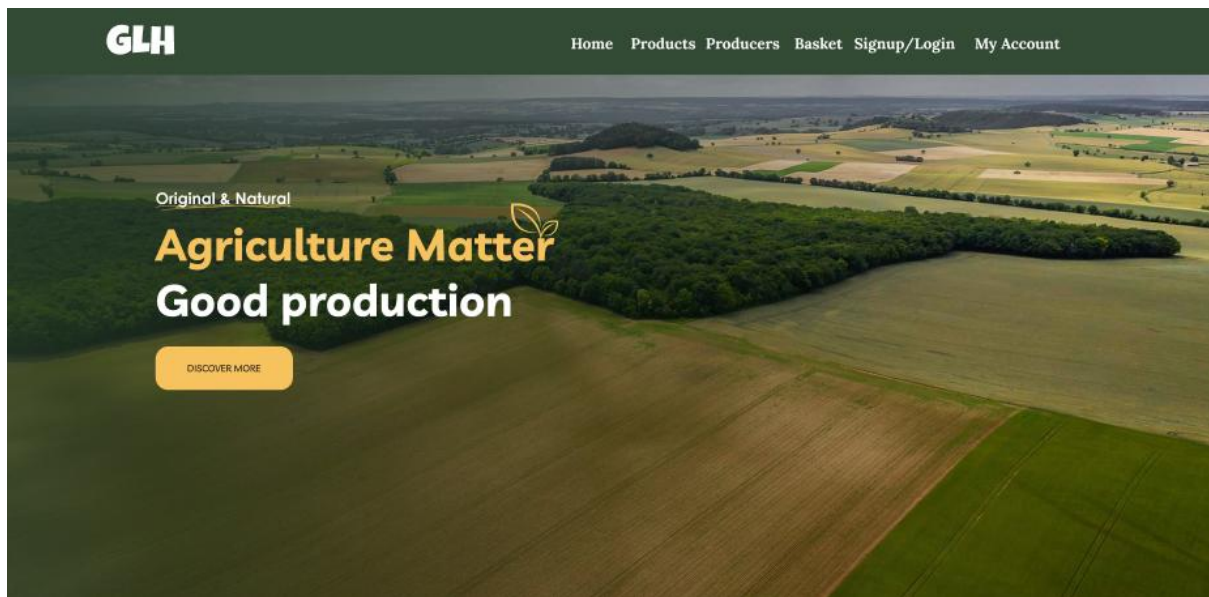
The freshest milk I've ever tasted. You can really tell the difference! - Sarah J.

Write a Review

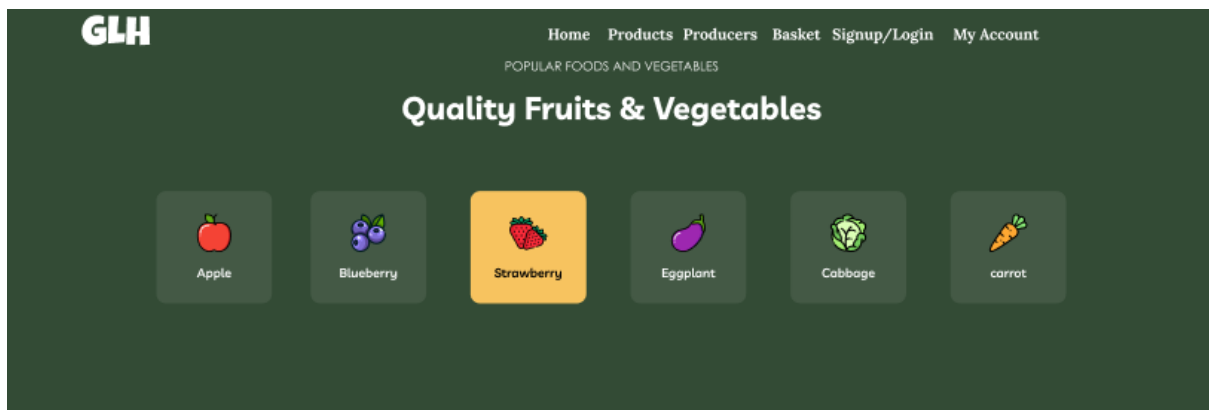
Submit Review

Visual/Interface Designs:

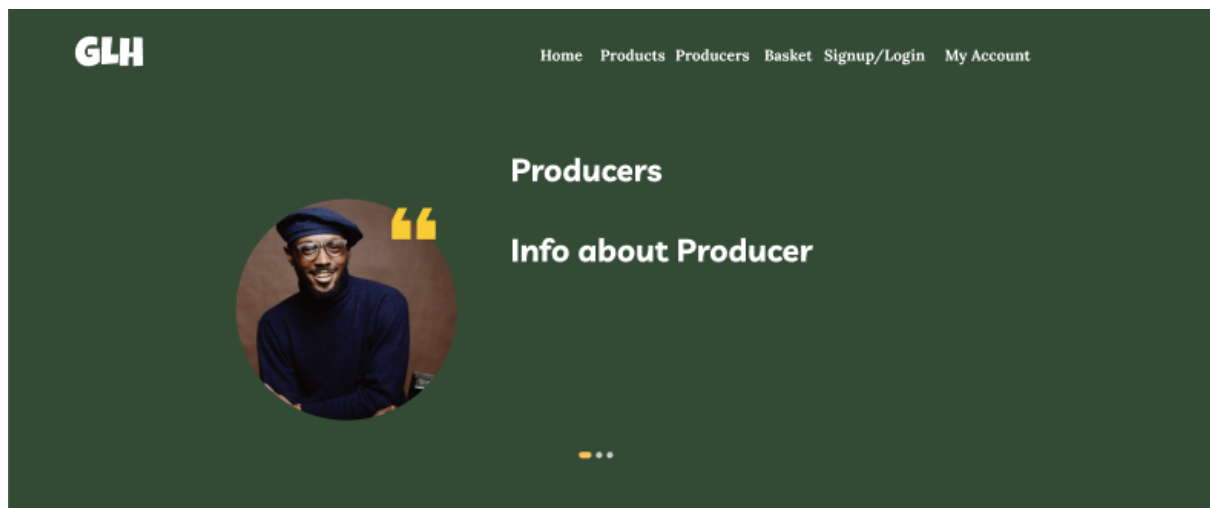
Home Page:



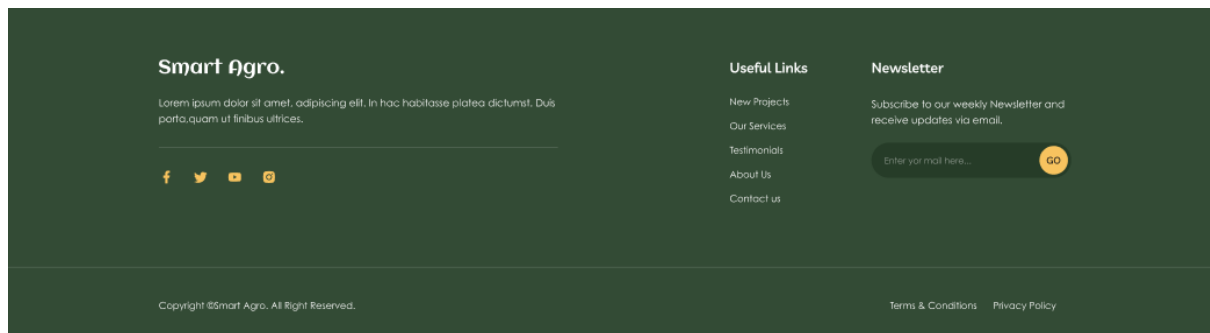
Products Page:



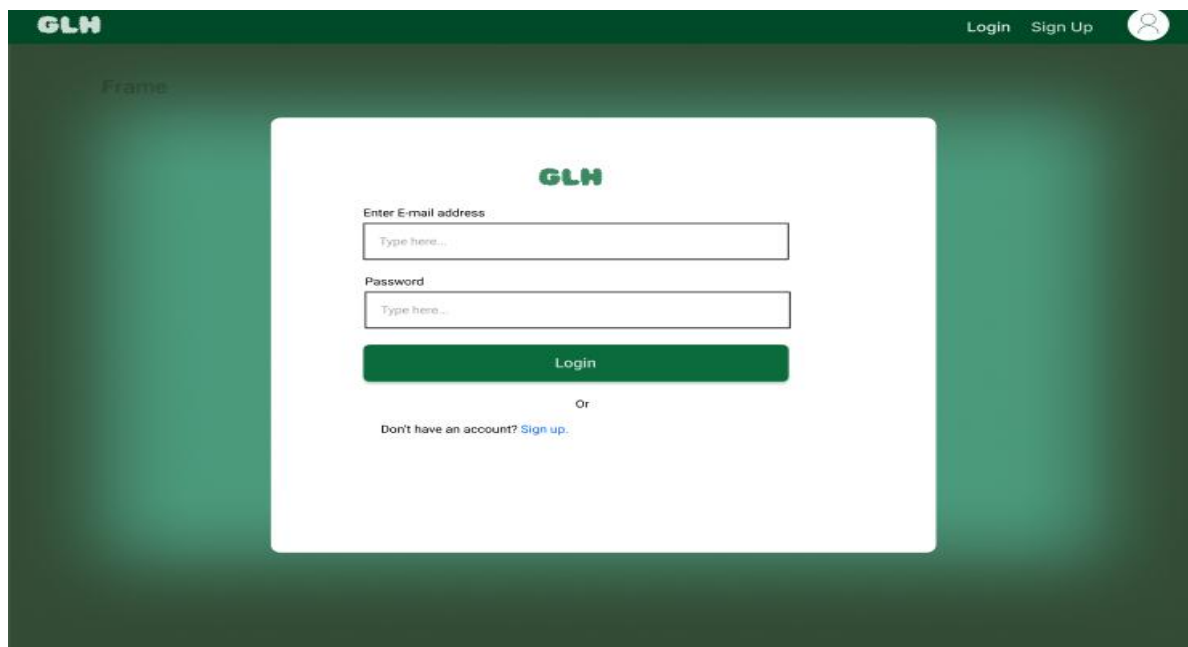
Producers:



Footers:



Login Page:



Sign Up Page:

GLH

Login Sign Up

GLH

Enter E-mail address

Type here...

Password

Type here...

Confirm Password

Type here...

Create account

Or

Already an member ? [Sign in.](#)

Algorithms Design

Flow Chart:

Pseudocode:

Customer Places Order with Stock Check + Loyalty Discount

FUNCTION placeOrder(customerID, basketItems, loyaltyPointsToUse, paymentDetails)

IF NOT isLoggedIn(customerID)

RETURN "Please log in first"

END IF

total = 0

FOR EACH item IN basketItems

stock = getStock(item.productID)

IF stock < item.quantity

RETURN "Not enough stock for " + item.name

END IF

```

    total = total + (item.price * item.quantity)
END FOR

discount = 0

IF loyaltyPointsToUse > 0

    pointsAvailable = getCustomerPoints(customerID)

    IF loyaltyPointsToUse > pointsAvailable

        RETURN "Not enough loyalty points"

    END IF

    discount = loyaltyPointsToUse * 0.05 // 1 point = 5p off

END IF

finalTotal = total - discount

IF processPayment(finalTotal, paymentDetails) == SUCCESS

    FOR EACH item IN basketItems

        reduceStock(item.productID, item.quantity)

    END FOR

    newPoints = floor(total / 10) // 1 point per £10

    updatePoints(customerID, pointsAvailable - loyaltyPointsToUse + newPoints)

    orderID = createOrder(customerID, basketItems, finalTotal)

    sendConfirmationEmail(customerID, orderID)

    RETURN "Order placed successfully"

ELSE

    RETURN "Payment failed"

END IF

END FUNCTION

```

Producer updates Stock in real time:

```

FUNCTION updateStock(producerID, productID, newQuantity)

    IF NOT isAuthorised(producerID, productID)

        RETURN "No permission for this product"
    
```



```

END IF

IF newQuantity < 0
    RETURN "Stock cannot be negative"
END IF

oldStock = getStock(productID)
saveStock(productID, newQuantity)
broadcastUpdateToCustomers(productID, newQuantity) // instant push to shop
pages

IF newQuantity < 10 AND oldStock >= 10
    sendLowStockEmail(producerID, productID, newQuantity)
END IF

RETURN "Stock updated successfully"
END FUNCTION

```

Customer Redeems Point at Checkout:

```

FUNCTION redeemLoyaltyAtCheckout(customerID, orderTotal, pointsToUse)
    availablePoints = getCustomerPoints(customerID)
    IF pointsToUse > availablePoints
        RETURN "Not enough points", orderTotal
    END IF

    discount = pointsToUse * 0.05 // example: 1 point = 5p
    newTotal = orderTotal - discount
    RETURN discount, newTotal
END FUNCTION

// After payment success

FUNCTION updatePointsPostPurchase(customerID, pointsUsed, orderTotal)
    earned = floor(orderTotal / 10) // 1 point per £10
    current = getCustomerPoints(customerID)
    updated = current - pointsUsed + earned

```

```
    saveCustomerPoints(customerID, updated)
```

```
END FUNCTION
```

Customer Tracks Order Status:

```
FUNCTION trackOrderStatus(customerID, orderID)
```

```
    IF NOT isLoggedIn(customerID)
```

```
        RETURN "Log in required"
```

```
    END IF
```

```
    order = getOrder(orderID)
```

```
    IF order.customerID != customerID
```

```
        RETURN "Not your order"
```

```
    END IF
```

```
    currentStatus = order.status
```

```
    lastViewed = getLastViewedStatus(customerID, orderID)
```

```
    IF currentStatus != lastViewed
```

```
        sendNotification(customerID, orderID, currentStatus)
```

```
        saveLastViewedStatus(customerID, orderID, currentStatus)
```

```
    END IF
```

```
    RETURN {
```

```
        currentStatus: currentStatus,
```

```
        history: order.statusHistory,
```

```
        estimatedTime: order.estimatedReadyTime
```

```
    }
```

```
END FUNCTION
```

GLH approves New Product:

```
FUNCTION submitNewProduct(producerID, name, price, description, photo)
```

```
    IF NOT isAuthorised(producerID)
```

```
        RETURN "Not authorised"
```

```
    END IF
```

```
product = {  
    producerID: producerID,  
    name: name,  
    price: price,  
    description: description,  
    photo: photo,  
    status: "Pending"  
}  
saveProduct(product)  
notifyAdminNewProduct(product.id)
```

```
    RETURN "Submitted for approval"
```

```
END FUNCTION
```

```
FUNCTION approveProduct(adminID, productID, approve)
```

```
    IF NOT isAdmin(adminID)
```

```
        RETURN "Not authorised"
```

```
    END IF
```

```
    product = getProduct(productID)
```

```
    IF product.status != "Pending"
```

```
        RETURN "Already processed"
```

```
    END IF
```

```
    IF approve == true
```

```
        product.status = "Live"
```

```
        saveProduct(product)
```

```
        sendApprovalEmail(product.producerID)
```

```
    ELSE
```

```
        product.status = "Rejected"
```

saveProduct(product)

sendRejectionEmail(product.producerID, reason)

END IF

RETURN "Product " + (approve ? "approved" : "rejected")

END FUNCTION

Data Dictionary:

Table Name	Field Name	Data Type	Size / Format	Description	Constraints / Rules
Customers	customerID	INTEGER	Auto-increment	Unique number for each customer	Primary Key, Auto-increment
Customers	name	VARCHAR	100	Full name of the customer	Not null
Customers	email	VARCHAR	150	Customer's email address (used for login)	Unique, Not null
Customers	password_hash	VARCHAR	255	Secure hashed version of the password	Not null
Customers	address	TEXT	-	Delivery address (house number, street, town, postcode)	-
Customers	phone	VARCHAR	20	Contact phone number	-

Customers	created_date	DATE	YYYY-MM-DD	Date the account was created	Not null, default current date
Products	productID	INTEGER	Auto-increment	Unique number for each product	Primary Key, Auto-increment
Products	name	VARCHAR	150	Name of the product (e.g. "Organic Carrots")	Not null
Products	price	DECIMAL	10,2	Price in pounds (e.g. 2.99)	Not null, >= 0
Products	stock	INTEGER	-	Current number of items in stock	Not null, >= 0
Products	producerID	INTEGER	-	Links to the producer who supplies this item	Foreign Key to Producers.producerID
Products	description	TEXT	-	Full details about the product (how grown, size, etc.)	-
Products	image_url	VARCHAR	255	Web link to the product photo	-
Products	category	VARCHAR	50	Type of product (e.g. Vegetables, Dairy, Meat, Bakery)	-
Products	harvest_date	DATE	YYYY-MM-DD	Date the item was	-

				harvested or made (for freshness)	
Products	status	ENUM	'active', 'pending', 'inactive'	Shows if product is live on shop or waiting for approval	Default 'pending'
Producers	producerID	INTEGER	Auto-increment	Unique number for each producer/farmer	Primary Key, Auto-increment
Producers	name	VARCHAR	100	Name of the producer or farm	Not null
Producers	email	VARCHAR	150	Producer's contact email	Unique, Not null
Producers	farm_description	TEXT	-	Short story about the farm and how they grow/make food	-
Producers	methods	TEXT	-	Growing methods (e.g. organic, free-range, no chemicals)	-
Producers	approved	BOOLEAN	-	Whether GLH staff has approved this producer	Default FALSE

Orders	orderID	INTEGER	Auto-increment	Unique number for each order	Primary Key, Auto-increment
Orders	customerID	INTEGER	-	Links to the customer who placed the order	Foreign Key to Customers.customerID
Orders	order_date	DATE	YYYY-MM-DD	Date the order was placed	Not null
Orders	order_time	TIME	HH:MM:SS	Time the order was placed	Not null
Orders	type	ENUM	'collection', 'delivery'	Whether customer collects or gets delivery	Not null
Orders	delivery_time	DATETIME	YYYY-MM-DD HH:MM	Chosen collection or delivery time slot	-
Orders	total_amount	DECIMAL	10,2	Final amount paid (after discount)	Not null, >= 0
Orders	status	VARCHAR	50	Current order status (Received, Preparing, Ready, Collected, Delivered)	Not null
Order_Items	orderItemID	INTEGER	Auto-increment	Unique number for each item in an order	Primary Key, Auto-increment

Order_Items	orderID	INTEGER	-	Links to the main order	Foreign Key to Orders.orderID
Order_Items	productID	INTEGER	-	Links to the product bought	Foreign Key to Products.productID
Order_Items	quantity	INTEGER	-	Number of this product ordered	Not null, > 0
Loyalty	loyaltyID	INTEGER	Auto-increment	Unique record for loyalty points	Primary Key, Auto-increment
Loyalty	customerID	INTEGER	-	Links to the customer	Foreign Key to Customers.customerID
Loyalty	points_balance	INTEGER	-	Current number of loyalty points	Not null, >= 0
Loyalty	last_updated	DATE TIME	YYYY-MM-DD HH:MM:SS	When the points were last changed	-

Test Strategy:

Date of test	Component to be tested	Type of test to be carried out	Prerequisites and dependencies
23 March 2026	Home page (navigation bar, footer, hero section, featured products grid, search bar)	Acceptance testing	Basic website layout and static pages created. Navigation bar and footer already built.
23 March 2026	Product catalogue/shop page (search bar, filters, product grid, pagination)	Acceptance testing	Home page complete. Some test products added to database.

24 March 2026	Single product detail page (image, price, stock, description, producer info, add to basket)	Acceptance testing	Shop page working. At least one product in database.
24 March 2026	Basket page (item list, quantity adjust, remove, total, proceed button)	Acceptance testing	Product detail page complete. Add to basket function working.
25 March 2026	Checkout page (delivery/collection choice, time slot picker, loyalty discount, payment)	Acceptance testing	Basket page complete. Test payment gateway in safe mode ready.
25 March 2026	Order confirmation and tracking page (status timeline, confirmation email)	Acceptance testing	Checkout complete. Test orders saved in database.
13 April 2026	Register page (form fields, validation, error messages, account creation)	Unit + Integration testing	Database ready. Email service test mode ready.
13 April 2026	Login page (authentication, error messages, forgot password link)	Unit + Integration testing	Register page tested. Test accounts created in database.
14 April 2026	My Account page (edit details, view order history, see loyalty points balance)	Acceptance testing	Login working. Test customer account created.
14 April 2026	Producer dashboard (update stock, edit products, view orders, low-stock alerts)	Acceptance testing	Producer test account ready. Products in database.
15 April 2026	Loyalty scheme (earn points on order, view balance, apply discount at checkout)	Integration testing	Checkout and order system complete. Loyalty rules tested.
15 April 2026	Accessibility panel (larger text, high contrast, screen reader mode, save settings)	Accessibility testing	All main pages built. Screen reader software ready for checking.
16 April 2026	Admin area (approve new producers, view sales)	Acceptance testing	Admin test account ready. Producers and orders in database.

	reports, update site news/offers)		
16 April 2026	Full customer order flow (browse → add to basket → checkout → track order)	System testing	All customer pages and database ready. Test payment in safe mode.